

Distance on the Coordinate Plane

Lesson 9-6

Name: _____

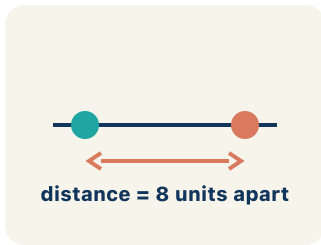
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

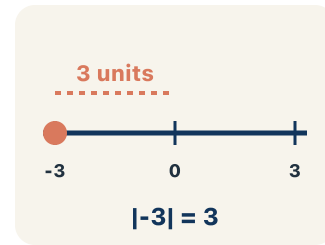
Picture first, then the word, then a plain-language meaning. Say each word out loud.



The distance from -3 to 4 on a number line is $|-3| + |4| = 3 + 4 = 7$ units

Distance

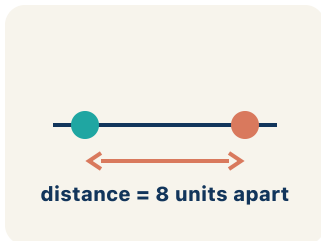
Write the definition:



$|-3| = 3$ and $|4| = 4$; to find distance: $|4 - (-3)| = 7$

Absolute value

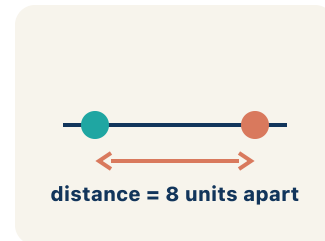
Write the definition:



From $(-2, 3)$ to $(5, 3)$: count from -2 to 5 = 7 units across

Horizontal distance

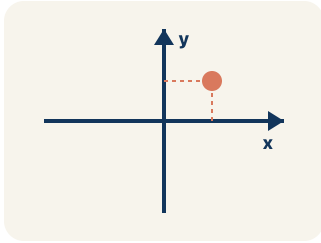
Write the definition:



From $(4, -1)$ to $(4, 6)$: count from -1 to 6 = 7 units up

Vertical distance

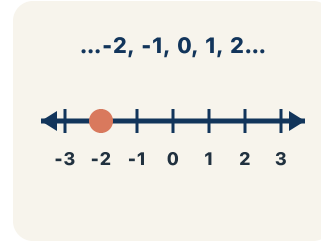
Write the definition:



Two number lines crossing at $(0, 0)$, creating four quadrants

Coordinate plane

Write the definition:



..., -3, -2, -1, 0, 1, 2, 3, ...

Integer

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the distance between two points on the coordinate plane using absolute value.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Distance

Absolute value

Horizontal distance

Vertical distance

Coordinate plane

Integer

1 How far apart two points are on the coordinate plane is the _____.

2 The distance of a number from zero, used to find distance between points, is its _____.

3 The distance between two points along the x-direction is the _____.

4 The distance between two points along the y-direction is the _____.

5 The grid made by the x-axis and y-axis is the _____.

6 A whole number or its opposite is an _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is the distance between $(-4, 3)$ and $(2, 3)$?

- A. 2 units
- B. 4 units
- C. 6 units
- D. 8 units

Step 1 Both points have $y = 3$, so this is a horizontal distance.

Step 2 $|2 - (-4)| = |2 + 4| = 6$ units.

 **Answer:** C. 6 units

 We do — together

Solve this one with your class using the same steps.

Problem: What is the distance between $(5, -2)$ and $(5, 4)$?

- A. 2 units
- B. 4 units
- C. 6 units
- D. 10 units

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: A rectangle has vertices at $(1, 1)$, $(5, 1)$, $(5, 4)$, $(1, 4)$. What is its perimeter?

- A. 14 units
- B. 12 units
- C. 20 units
- D. 8 units

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. The Park is at $(-4, 5)$ and City Hall is at $(3, 5)$. How many blocks apart are they?

- A. 1 block
- B. 7 blocks
- C. 12 blocks
- D. 5 blocks

Show your work:

2. The Café is at $(-2, 1)$ and the Gym is at $(-2, -4)$. How many blocks apart are they?

- A. 3 blocks
- B. 5 blocks
- C. 6 blocks
- D. 2 blocks

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A rectangle has vertices at $(-4, -2)$, $(3, -2)$, $(3, 5)$, and $(-4, 5)$. Find the length, width, perimeter, and area of the rectangle. Show your work using absolute value.

Sentence starter: The width is $|______ - (______)| = ______ \text{ units}$. The height is $|______ - (______)| = ______ \text{ units}$. The perimeter is $2(______ + ______) = ______ \text{ units}$. The area is $______ \times ______ = ______ \text{ square units}$.

Show your work:

Reflect — Exit Ticket

What is the distance between the points $(3, -2)$ and $(3, 5)$?

- A. 3 units
- B. 5 units
- C. 7 units
- D. 9 units

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Distance
2. **Notes 2:** Absolute value
3. **Notes 3:** Horizontal distance
4. **Notes 4:** Vertical distance
5. **Notes 5:** Coordinate plane
6. **Notes 6:** Integer
7. **We Try (notes):** C. 6 units — *Both points have $x = 5$, so this is a vertical distance. $|4 - (-2)| = |4 + 2| = 6$ units.*
8. **You Try (notes):** A. 14 units — *Width = $5 - 1 = 4$. Height = $4 - 1 = 3$. Perimeter = $2(4 + 3) = 14$.*
9. **Try It 1:** B. 7 blocks — *Same $y = 5$, so the segment is horizontal. The points are on opposite sides of zero, so add the absolute values: $|-4| + |3| = 4 + 3 = 7$ blocks.*
10. **Try It 2:** B. 5 blocks — *Same $x = -2$, so the segment is vertical. The points are on opposite sides of zero: $|1| + |-4| = 1 + 4 = 5$ blocks.*
11. **Exit Ticket:** C. 7 units — *Both points share $x = 3$, so this is a vertical distance. $|5 - (-2)| = |5 + 2| = 7$ units.*